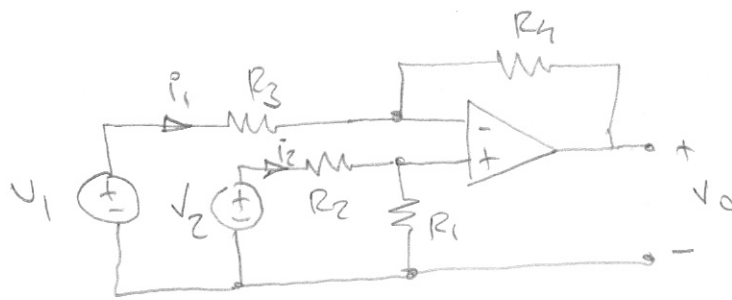


8.20



$$R_1 = R_2 = R_3 = R_4 = R$$

$$V_1 = 4V \quad V_2 = 5V$$

BESTIMM V_0

Lösung: Ansatz Kirchhoff

Knoten

$$\begin{cases} -V_1 + R_3 i_1 + R_1 i_2 = 0 & (1) \\ -V_2 + R_2 i_2 + R_1 i_2 = 0 & (2) \\ -V_0 - R_4 i_1 + R_1 i_2 = 0 & (3) \end{cases}$$

$$(2) \text{ zur } V_2 = (R_1 + R_2) i_2 \Rightarrow i_2 = \frac{V_2}{R_1 + R_2} \quad (4)$$

$$(1) \text{ zur } -V_1 + R_3 i_1 + R_1 \frac{V_2}{R_1 + R_2} = 0$$

$$i_1 = \frac{1}{R_3} \left(V_1 - \frac{R_1 V_2}{R_1 + R_2} \right) \quad (5)$$

(5) oder (4) i (3) zur

$$V_0 = R_1 \frac{V_2}{R_1 + R_2} - R_4 \frac{1}{R_3} \left(V_1 - \frac{R_1 V_2}{R_1 + R_2} \right)$$

MEN ALLE RESISTANZEN AN N LIA, DETA ZUR

$$V_0 = \frac{R}{2R} V_2 - \frac{R}{R} \left(V_1 - \frac{R}{2R} V_2 \right) =$$

$$= \frac{1}{2} V_2 - V_1 + \frac{1}{2} V_2 = \underline{\underline{V_2 - V_1 = 1V}}$$

$$\underline{\text{SVAR:}} \quad V_0 = 1 \text{ Volt}$$